



ADMISSION NUMBER

School of Computer Applications and Technology
Mid Term Examination - Nov 2025

Masters of Computer Applications

Duration : 90 Minutes
Max Marks : 50

Sem I - E1PA101T - Computational Mathematics and Statistics

General Instructions

Answer to the specific question asked

Draw neat, labelled diagrams wherever necessary

Approved data hand books are allowed subject to verification by the Invigilator

1) Define Upper triangular matrix. Give an example of Upper triangular matrix of order 4×4 . Find trace of given Upper triangular matrix. K1 (3)

2) Define mean and find the mean of values 25, 20, 15, 35, 18, 40, 50. K1 (3)

3) Find inverse of the given matrix using Gauss Jordan method. $A = \begin{bmatrix} 1 & 1 \\ 1 & 3 \end{bmatrix}$ K2 (4)

4) Calculate mean and median of the data {14, 16, 16, 16, 17, 16, 18} K2 (4)

5) Solve by Gauss elimination method:
 $x - 2y + z = 0$
 $2x + y - 3z = 5$
 $4x - 7y + z = -1$ K3 (5)

6) Find the median. K3 (5)

Marks	No. of students
5-15	10
15-25	20
25-35	25
35-45	20
45-55	15
55-65	10

7) $A = \begin{bmatrix} 5 & 8 & 16 \\ 4 & 1 & 8 \\ -4 & -4 & -11 \end{bmatrix}$ Show A is diagonalizable K4 (8)

8) Obtain the line of best fit to the following data is given by

X	Y
6	5
7	5
7	4
8	5
8	4
8	3
9	4
9	3
10	3

K4 (8)

9) Find algebraic multiplicity and Geometric multiplicity of the matrix

$$A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 4 \\ 0 & 4 & 9 \end{bmatrix}$$

K5 (10)

OR

Calculate the value of the correlation coefficient for the given below data

K5 (10)

x	64	66	67	67	68	69	70	71
y	60	68	65	68	72	72	69	70